A REGULAR RING OF FINITE TYPE OVER A FIELD IS FORMALLY SMOOTH AFTER INVERTING ONE ELEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Formal smoothness and the Jacobian criterion for a finite type algebra over a field. used in both cases, proof lines 9894-9951. Supplies the criterion that has to be satisfied after inverting the element.
- Regularity of a finite type algebra over a field and its consequences for the module of differentials. used at proof lines 9894-9920. Provides the algebraic input that makes the differential criterion applicable.
- p-basis theory in characteristic p, developed in the same chapter. the statement counted by the sibling card S1-algebra-p-basis at 11362-11415. Supplies the characteristic p case of the proof.

Source reference. Original lemma 07EL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9451–9451. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07EI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9448–9448. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9449–9449. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07E5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9435–9435. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9446–9446. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9441–9441. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00RO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1010–1010. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Theorem 0.2. *Let k be a field. Let $(A, \mathfrak{m}, K)$ be a Noetherian local k-algebra. If the characteristic of k is zero then the following are equivalent*

1. *A is a regular local ring, and*
2. *$k \to A$ is formally smooth in the $\mathfrak{m}$-adic topology.*

If the characteristic of k is $p > 0$ then the following are equivalent

1. *A is geometrically regular over k,*
2. *$k \to A$ is formally smooth in the $\mathfrak{m}$-adic topology.*
3. *for all $k \subset k' \subset k^{1/p}$ finite over k the ring $A \otimes_k k'$ is regular,*
4. *A is regular and the canonical map $H_1(L_{K/k}) \to \mathfrak{m}/\mathfrak{m}^2$ is injective, and*
5. *A is regular and the map $\Omega_{k/\mathbf{F}_p} \otimes_k K \to \Omega_{A/\mathbf{F}_p} \otimes_A K$ is injective.*

Proof. If the characteristic of k is zero, then the equivalence of (1) and (2) follows from Lemmas 07EI and 07EJ.

If the characteristic of k is $p > 0$, then it follows from Proposition 07E5 that (1), (3), (4), and (5) are equivalent. Assume (2) holds. By Lemma 07EG we see that $k' \to A' = A \otimes_k k'$ is formally smooth for the $\mathfrak{m}' = \mathfrak{m}A'$-adic topology. Hence if $k \subset k'$ is finite purely inseparable, then A' is a regular local ring by Lemma 07EI. Thus we see that (1) holds.

Finally, we will prove that (5) implies (2). Choose a solid diagram

$$\begin{array}{ccc} A & \longrightarrow & B/J \\ {\scriptstyle i}\uparrow & \overset{\bar\psi}{\searrow} & \uparrow{\scriptstyle \pi} \\ k & \xrightarrow{\varphi} & B \end{array}$$

as in Definition 07EB. As $J^2 = 0$ we see that J has a canonical B/J module structure and via $\bar\psi$ an A-module structure. As $\bar\psi$ is continuous for the $\mathfrak{m}$-adic topology we see that $\mathfrak{m}^n J = 0$ for some n. Hence we can filter J by B/J-submodules $0 \subset J_1 \subset J_2 \subset \ldots \subset J_n = J$ such that each quotient J_{t+1}/J_t is annihilated by $\mathfrak{m}$. Considering the sequence of ring maps $B \to B/J_1 \to B/J_2 \to \ldots \to B/J$ we see that it suffices to prove the existence of the dotted arrow when J is annihilated by $\mathfrak{m}$, i.e., when J is a K-vector space.

Assume given a diagram as above such that J is annihilated by $\mathfrak{m}$. By Lemma 07EJ we see that $\mathbf{F}_p \to A$ is formally smooth in the $\mathfrak{m}$-adic topology. Hence we

can find a ring map $\psi : A \to B$ such that $\pi \circ \psi = \bar{\psi}$. Then $\psi \circ i, \varphi : k \to B$ are two maps whose compositions with π are equal. Hence $D = \psi \circ i - \varphi : k \to J$ is a derivation. By Algebra, Lemma 00RO we can write $D = \xi \circ \mathrm{d}$ for some k-linear map $\xi : \Omega_{k/\mathbf{F}_p} \to J$. Using the K-vector space structure on J we extend ξ to a K-linear map $\xi' : \Omega_{k/\mathbf{F}_p} \otimes_k K \to J$. Using (5) we can find a K-linear map $\xi'' : \Omega_{A/\mathbf{F}_p} \otimes_A K$ whose restriction to $\Omega_{k/\mathbf{F}_p} \otimes_k K$ is ξ'. Write

$$D' : A \xrightarrow{\mathrm{d}} \Omega_{A/\mathbf{F}_p} \to \Omega_{A/\mathbf{F}_p} \otimes_A K \xrightarrow{\xi''} J.$$

Finally, set $\psi' = \psi - D' : A \to B$. The reader verifies that ψ' is a ring map such that $\pi \circ \psi' = \bar{\psi}$ and such that $\psi' \circ i = \varphi$ as desired. □